WEEK 2 · LESSON 12 OF 24 · HANDOUT

Week 2 Project: Plan a Machine-Learning Project

Study time: **about 60 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=R9OHn5ZF4Uo — open the lesson online to watch inside the course.

Your goal in this lesson: Write a complete, realistic plan for a machine-learning project, using the workflow, data and learning-type ideas from Week 2.

About this project

Re-watch this short video on how machines learn, then write a **project plan** of 2–3 pages for one machine-learning idea that could help people in Rwanda or your own community. You will not write code. You will think like the person who leads the project — the most valuable skill in real AI teams.

Step 1 — The problem

- Who has the problem? What decision do they make today, and how?
- What will the model predict or find?
- **Success measure:** write it as a number (for example, "find 80% of fraud cases while wrongly flagging fewer than 5% of normal ones").
- Could a simple rule solve it without AI? Explain why ML is (or is not) needed.

Step 2 — Learning type and problem type

- Supervised, unsupervised or reinforcement learning? Why?
- Classification, regression, clustering, anomaly detection…? Why?

Step 3 — The data

Make a table of the features and label:

Feature / label	Type (number, category, text, image…)	Where it comes from	Possible quality problem

Then answer: - Is the data **representative** of everyone the model will serve? - Is there any risk of **leakage**? - Do you have **permission** to use this data? Does it contain personal information?

Step 4 — Method

Name one simple **baseline** method (for example logistic regression or a decision tree) and one stronger method you might try later (for example a random forest or a neural network). Explain the trade-off between them.

Step 5 — Deployment and monitoring

- How will the user actually receive the result (SMS, app, dashboard, report)?
- What could cause **drift** in your problem?

- How often will you check and retrain?

Step 6 — Risks

Name two things that could go wrong (wrong predictions, unfair results for one group, privacy problems) and how you would reduce each one.

Example ideas

- Predict which Grade 6 pupils may drop out, so schools can support them early.
- Detect unusual electricity use that may mean a faulty meter.
- Group small shops by sales patterns to design better supplier credit.
- Predict demand for buses on each route by hour.

Checklist

- Every step is answered in clear, simple sentences.
- The success measure is a number.
- The data table has at least four features.
- Risks are honest and specific.

Remember this: A strong plan names the decision, a numeric success measure, the right learning type, representative data with permission, a simple baseline first, and a plan for drift and risk.

Name: _____ Date: _____

Activity

Task 1

Complete the six steps for your idea. Share it with a classmate, a teacher or a professional in that field and ask: *"Would this help you? What did I miss?"* Write down their feedback and one change you made. Keep this plan — it feeds into your final project.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. How should the success measure in your plan be written?
   A. As a feeling, like 'works well'
   B. It is not needed
   C. As a number, such as 'find 80% of fraud cases'
   D. As the model's name
2. Why start with a simple baseline method?
   A. Simple methods are always the most accurate
   B. To avoid collecting data
   C. It gives a reference point to judge whether a complex method is worth it
   D. Because complex methods are illegal
3. Which question checks if your data is representative?
   A. Does the data include everyone the model will serve?
   B. Is the computer fast?
   C. Is the data in English?
   D. Is the file name short?
4. Predicting which pupils may drop out using past records with known outcomes is…
   A. Supervised classification
   B. Reinforcement learning
   C. Generation
   D. Unsupervised clustering
5. Why ask 'could a simple rule solve this without AI?'
   A. Because ML is always wrong
   B. Because ML costs time and money and is only worth it when rules are not enough
   C. To make the plan shorter
   D. Because rules are illegal
6. Which belongs in the deployment step?
   A. The founder's biography
   B. The colour of the model
   C. The number of slides
   D. How the user will receive the result and how drift will be checked

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